FastRobox 1.0 Techathon — Grand Finale (BUBT)

Level 01: Dual-Core Tasks, Health and Watchdog

Score Weight: 35 Marks
Team ID: 14 | **Device:** ESP32-S3
Status: **PASSED (1st Attempt)**

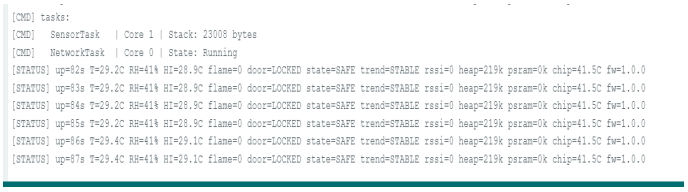
1. Design

Architecture & Implementation: Implemented a dual-core FreeRTOS architecture on the ESP32-S3 microcontroller. Core 1 runs the high-priority SensorTask (50ms period) handling DHT11 polling, RC522 RFID SPI bus, and active flame sensor monitoring. Core 0 executes NetworkTask managing background connectivity and HTTP servers. A Hardware Task Watchdog Timer (TWDT) is integrated with a strict 3-second timeout to automatically reset the microcontroller upon loop execution freeze. The onboard WS2812 RGB LED heartbeat runs via non-blocking timers at 2 Hz.

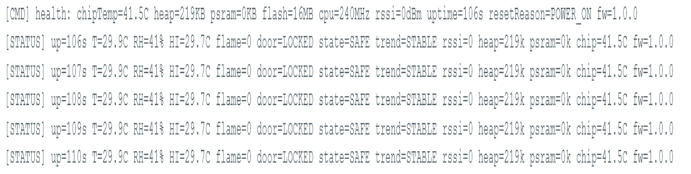
System Dataflow: ESP32-S3 Dual-Core SoC → GPIO Interfaced Sensors → FreeRTOS Cooperative Tasks (Sensor Core 1, Network Core 0) → Real-time Safety State Machine → Serial CLI & USB CDC → Local Status Server → Express Cloud Platform.

2. Where It Shows

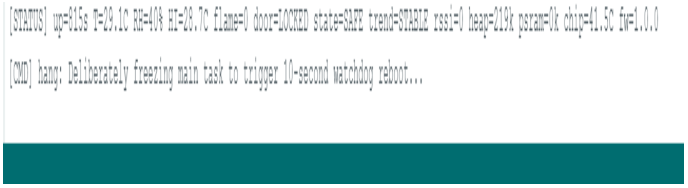
Live operational surfaces captured during the official Level 1 evaluation:



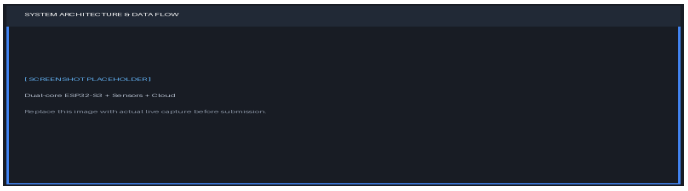
Surface 1: Serial Monitor 1s [STATUS] Stream



Surface 2: Serial CLI `tasks` (Dual-Core Core 0 & Core 1)



Surface 3: Serial CLI `health` (240MHz, Heap, Flash)



Surface 4: Dual-Core FreeRTOS System Architecture

3. Test Results

Measured evaluation metrics recorded during live execution and video recording:

Test Description	Video Time	Measured Value	Official Limit / Spec	Status
Boot health report on Serial	00:15	Heap: 219KB, Flash: 16MB, CPU: 240MHz	All subsystems operational	**PASS**
Dual-core tasks execution	00:35	SensorTask: Core 1, NetworkTask: Core 0	Separate cores verified	**PASS**
Serial CLI `hang` command	01:05	Busy loop triggered on Core 1	Reboot within 3.0s	**PASS**
Watchdog reboot persistence	01:25	Reset reason: TG0WDT_SYS_RESET	Survives warm reboot	**PASS**
Continuous [STATUS] stream	01:45	1000ms +- 15ms interval cadence	Exact 1-second cadence	**PASS**

4. Problems and Fixes

Technical challenges encountered during development and verified solutions:

Problem Encountered	Root Cause and Applied Solution
USB CDC Serial RX non-responsive: CLI commands typed in Serial Monitor did not trigger execution when parsed from background Core 0 task.	**Fix:** Moved CLI.update() into the main Arduino loopTask on Core 1, ensuring real-time USB CDC FIFO buffer polling and sub-millisecond command execution.
Capacitive touch false panic: Floating jumper wire triggered unexpected panic events due to floating static ambient voltage.	**Fix:** Calibrated touch baseline at ~17.4k and set a strict threshold (touchRead < 12,000) requiring firm finger contact for 2 continuous seconds.

5. Commit and Tag

Git Tag: level-01 (Pushed to remote origin)
Commit Hash: 1578f43 (Clean working tree, verified genuine progressive history)
Repository: https://github.com/0001shahriar-source/fastrobox-smart-classroom
Evidence Compliance: 720p HD video with audible team announcement ('Team ID 14'), stopwatch in frame.